

MONTIJO AB, Portugal

WMO: 085340

Lat: 38.7139N

Lon: 9.0245W

Elev: 11

StdP: 101.19

Time Zone: 0.00 (EUW)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.1	(c) 4.8	(d) -0.5	(e) 3.6	(f) 6.8	(g) 1.2	(h) 4.1	(i) 7.7	(j) 10.9	(k) 14.2	(l) 9.7	(m) 13.0	(n) 2.6	(o) 30	(p) 0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.5	(c) 33.2	(d) 21.0	(e) 31.5	(f) 20.7	(g) 29.9	(h) 20.4	(i) 22.3	(j) 30.2	(k) 21.6	(l) 28.8	(m) 21.0	(n) 27.6	(o) 4.4	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.1	(b) 14.8	(c) 23.5	(d) 19.4	(e) 14.2	(f) 23.4	(g) 18.9	(h) 13.8	(i) 23.3	(j) 65.1	(k) 29.8	(l) 62.7	(m) 29.0	(n) 60.7	(o) 28.0	(p) 26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.5	(b) 8.3	(c) 7.4	(e) 0.6	(f) 37.9	(g) 1.6	(h) 2.6	(i) -0.6	(j) 39.8	(k) -1.5	(l) 41.3	(m) -2.4	(n) 42.8	(o) -3.6	(p) 44.7		
			DB	WB	0.0	24.1	1.6	0.9	-1.1	24.7	-2.0	25.2	-2.9	25.7	-4.0	26.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	11.1	12.0	14.1	16.0	18.7	21.4	23.2	23.7	22.0	19.1	14.5	11.6
	DBStd	5.02	2.43	2.12	2.12	2.42	2.71	2.48	2.03	1.90	1.88	2.38	2.58	2.53
	HDD10.0	36	16	7	1	0	0	0	0	0	0	0	1	12
	HDD18.3	983	223	176	132	77	27	2	0	0	0	19	118	208
	CDD10.0	2708	51	64	128	181	269	341	409	424	361	283	134	62
	CDD18.3	613	0	0	1	8	38	93	151	166	111	43	2	0
	CDH23.3	4141	0	0	14	67	312	616	1046	1236	654	196	1	0
	CDH26.7	1381	0	0	1	7	95	214	368	464	190	43	0	0
Wind	WSAvg	3.8	3.5	3.7	3.9	4.0	4.0	4.1	4.2	4.0	3.5	3.5	3.7	3.6
Precipitation	PrecAvg	735	105	92	72	62	47	18	3	5	30	94	121	101
	PrecMax	1189	353	298	267	163	156	98	43	58	129	262	373	383
	PrecMin	416	0	0	1	5	0	0	0	0	0	0	2	4
	PrecStd	206	76	67	59	40	40	22	7	11	32	74	86	91
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.9	25.1	27.9	32.9	34.8	36.1	36.2	33.1	30.9	22.9	19.0
		MCWB	14.7	14.4	15.7	18.8	20.5	21.3	22.3	21.5	20.1	19.0	17.1	16.6
	2%	DB	16.8	18.1	22.2	25.2	29.1	31.6	32.9	33.0	30.9	27.4	21.0	17.9
		MCWB	14.4	13.7	15.0	17.4	19.2	21.1	21.4	21.1	20.1	19.0	17.0	15.7
	5%	DB	16.0	17.0	20.1	23.1	26.8	29.0	30.2	31.0	28.9	25.2	19.8	16.9
		MCWB	13.9	13.7	14.7	16.6	18.5	20.1	20.7	21.0	19.9	18.5	16.2	14.9
	10%	DB	15.1	16.0	18.2	21.0	24.2	26.8	28.2	29.1	27.0	23.8	18.6	15.9
		MCWB	13.2	13.2	13.9	15.9	17.7	19.4	20.2	20.3	19.5	18.3	15.7	14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	16.0	17.9	19.6	21.6	23.0	23.2	23.0	22.2	21.6	19.5	17.3
		MCDB	16.8	17.5	22.6	26.3	28.3	32.3	34.0	31.3	28.5	25.8	21.0	18.4
	2%	WB	15.2	15.2	16.2	18.1	20.2	21.7	22.1	22.2	21.3	20.7	18.2	16.1
		MCDB	16.2	16.6	20.0	23.7	26.8	29.4	30.5	29.9	27.3	23.9	19.9	17.4
	5%	WB	14.5	14.5	15.4	17.1	19.1	20.7	21.3	21.5	20.7	20.0	17.2	15.2
		MCDB	15.5	16.2	18.5	21.9	25.5	27.5	28.6	28.8	26.4	23.0	18.8	16.6
	10%	WB	13.7	13.7	14.6	16.1	18.1	19.8	20.6	20.8	20.1	19.2	16.3	14.3
		MCDB	14.8	15.7	17.5	20.2	23.5	25.9	27.3	27.7	25.4	22.4	18.0	15.7
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.9	8.6	8.8	9.8	10.1	10.7	11.5	10.6	9.3	7.5	7.3
		MCDBR	7.4	9.2	12.3	12.7	14.2	14.6	14.4	14.7	13.9	12.4	8.5	7.3
		MCWBR	5.1	5.8	6.4	6.5	6.4	6.2	5.7	5.7	5.5	5.7	5.1	5.2
	5% WB	MCDBR	5.7	7.5	9.8	11.3	12.5	12.8	12.4	12.5	11.2	9.0	6.5	6.4
		MCWBR	4.6	5.5	5.9	6.1	6.3	5.9	5.5	5.4	5.1	4.9	4.6	5.1
Clear-Sky Solar Irradiance	taub	0.330	0.344	0.374	0.390	0.397	0.402	0.386	0.399	0.398	0.366	0.341	0.331	
	taud	2.506	2.463	2.388	2.340	2.348	2.356	2.391	2.363	2.379	2.490	2.516	2.505	
	Ebn at Noon	846	880	884	888	884	876	888	868	847	844	826	816	
	Edh at Noon	78	93	110	123	124	123	118	118	110	89	77	74	
All-Sky Solar Radiation	RadAvg	2.25	3.25	4.40	5.65	6.78	7.38	7.53	6.82	5.33	3.66	2.51	2.03	
	RadStd	0.28	0.38	0.47	0.47	0.47	0.39	0.29	0.26	0.34	0.31	0.31	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page